

ML Model Selection Guide

Choose the right model based on your data type and goal

1. Structured Data

Tables, Numbers, CSV

Predict a number (regression)



Linear Regression / Ridge / Lasso

Classify categories



XGBoost / LightGBM

Find hidden groups (clustering)



K-Means / DBSCAN

Detect outliers



Isolation Forest

Sequential over time (forecasting)



LSTM / Prophet / TimesGPT

2. Unstructured Data

Text, Images, Audio, Video

Generate text



LLM - GPT / Llama / Gemini

Classify text



BERT / RoBERTa

Classify images



Vision Transformer (ViT)

Generate images



Diffusion Model (Stable Diffusion)

Understand video



Video Transformer

Speech / Audio processing



Whisper / wav2vec

3. Interaction / Behavior Data

User clicks, purchases, browsing history

Recommend items



Collaborative Filtering / Two-Tower

Learn from rewards



Reinforcement Learning / PPO

4. Graph / Relational Data

Social networks, knowledge graphs, molecular structures

Node classification / Link prediction



Graph Neural Network (GNN)

Quick Reference Summary

Data Type	Goal	Recommended Model
Structured	Predict number	Linear Regression
Structured	Classify	XGBoost / LightGBM
Structured	Cluster	K-Means
Structured	Outliers	Isolation Forest
Structured	Time series	LSTM / Prophet
Unstructured	Generate text	GPT / Llama / Gemini
Unstructured	Classify text	BERT / RoBERTa
Unstructured	Classify images	Vision Transformer
Unstructured	Generate images	Diffusion Model
Unstructured	Video	Video Transformer
Unstructured	Audio/Speech	Whisper / wav2vec
Interaction	Recommend	Collab Filtering / Two-Tower
Interaction	RL	PPO / DQN
Graph	Relational	GNN